

ACM Observability Runbook

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Objective

This document walks through connecting a second OpenShift cluster to Red Hat Advanced Cluster Management and using it to deploy and monitor a virtual machine on that cluster, this covers:

- Enable observability for the imported cluster in ACM
- Import one cluster in ACM
- Deploy virtualization on the secondary cluster
- Deploy a vm from main hub cluster using gitops

Prerequisites

- Two clusters
- ACM deployed and configured in a cluster
- One github repo for hosting vm.yaml
- Admin access for both clusters

Procedure

1.1 Import the new cluster into ACM

ACM can only manage and observe clusters it knows about, so the second cluster needs to be registered with the hub first.

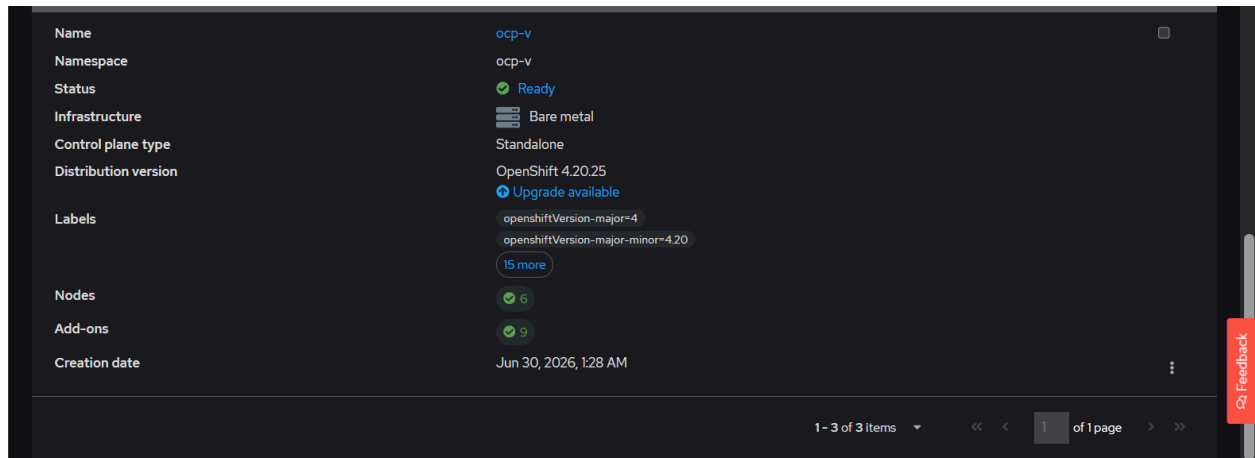
- In the ACM web console, click Import Cluster and fill in the details as shown below

The screenshot shows the Red Hat OpenShift console interface. The top navigation bar includes the Red Hat OpenShift logo, a dropdown for 'All Clusters', and user information 'kube:admin'. A blue banner at the top states: 'You are logged in as a temporary administrative user. Update the cluster OAuth configuration to allow others to log in.' The left sidebar contains navigation links: Home, Search, Infrastructure (highlighted), Applications, Governance, and Credentials. The main content area is titled 'Import an existing cluster' with a 'YAML' toggle. A progress indicator on the left shows three steps: 1. Details (active), 2. Automation, and 3. Review. The 'Details' section contains the following fields: 'Name' (text input with 'ocp-v'), 'Cluster set' (dropdown menu with 'default' selected and a 'Manage cluster sets' link), 'Additional labels' (text input with placeholder 'Enter key=value, then press return, space, or comma'), and 'Import mode' (dropdown menu with 'Run import commands manually' selected). Below these fields, a note states: 'You are provided a command to run after you submit the form.' At the bottom of the form are three buttons: 'Next' (highlighted in blue), 'Back', and 'Cancel'. A red 'Feedback' button is visible on the right edge.

Leave the remaining settings at their defaults and click Generate Command

This screenshot shows the same 'Import an existing cluster' form, but now in the 'Review' step. The progress indicator on the left shows: 1. Details, 2. Automation, and 3. Review (active). The 'Details' section now displays the values entered in the previous step: 'Name' is 'ocp-v2', 'Cluster set' is 'default', 'Additional labels' is empty, 'Import mode' is 'Run import commands manually', and 'Namespace' and 'Credential' are empty. At the bottom of the form, the 'Generate command' button is highlighted with a red rectangle, alongside 'Back' and 'Cancel' buttons. The 'Feedback' button remains on the right edge.

Copy that command and run it from the bastion node of the second cluster. It typically takes 10–20 seconds for the cluster to register with ACM



1.2 Deploy Virtualization operator in ocp-v cluster

Before a VM can run on this cluster, OpenShift Virtualization (CNV) needs to be installed. The steps below create the namespace, subscribe to the operator, and then bring up the HyperConverged custom resource that activates it.

- The operator can be installed from the UI or the CLI; this guide uses the CLI for repeatability

```
apiVersion: v1
kind: Namespace
metadata:
  name: openshift-cnv
---
apiVersion: operators.coreos.com/v1
kind: OperatorGroup
metadata:
  name: kubevirt-hyperconverged-group
  namespace: openshift-cnv
spec:
  targetNamespaces:
    - openshift-cnv
---
apiVersion: operators.coreos.com/v1alpha1
kind: Subscription
metadata:
  name: hco-operatorhub
  namespace: openshift-cnv
spec:
  source: redhat-operators
```

```
sourceNamespace: openshift-marketplace
name: kubevirt-hyperconverged
channel: stable
```

Once the subscription above is applied and the operator finishes installing, deploy the HyperConverged custom resource below to activate it

```
apiVersion: hco.kubevirt.io/v1beta1
kind: HyperConverged
metadata:
  name: kubevirt-hyperconverged
  namespace: openshift-cnv
spec: {}
```

1.3 Enable observability in ACM

ACM's observability service stores its metrics in an S3-compatible object bucket rather than local disk. The steps below provision that bucket through NooBaa, capture its access credentials, and point ACM at it.

- Create ObjectBucketClaim

```
apiVersion: objectbucket.io/v1alpha1
kind: ObjectBucketClaim
metadata:
  name: thanos-obc
  namespace: open-cluster-management-observability
spec:
  generateBucketName: thanos-bucket
  storageClassName: openshift-storage.noobaa.io
```

- Pull the bucket name, endpoint, and access keys from the claim using the commands below

```
oc get obc thanos-obc -n open-cluster-management-observability -o yaml
oc get route -n openshift-storage s3 -o jsonpath='{.spec.host}'
oc get secret thanos-obc -n open-cluster-management-observability -o
jsonpath='{.data.AWS_ACCESS_KEY_ID}' | base64 -d

oc get secret thanos-obc -n open-cluster-management-observability -o
jsonpath='{.data.AWS_SECRET_ACCESS_KEY}' | base64 -d
```

- Create the storage secret below, substituting the bucket name, endpoint, and keys gathered above

```

apiVersion: v1
kind: Secret
metadata:
  name: thanos-object-storage
  namespace: open-cluster-management-observability
type: Opaque
stringData:
  thanos.yaml: |
    type: s3
    config:
      bucket: <BUCKET_NAME>
      endpoint: <ENDPOINT>
      insecure: true
      access_key: <ACCESS_KEY>
      secret_key: <SECRET_KEY>

```

- Finally, create the MultiClusterObservability resource, referencing that secret as its storage backend

```

apiVersion: observability.open-cluster-management.io/v1beta2
kind: MultiClusterObservability
metadata:
  name: observability
spec:
  observabilityAddonSpec: {}
  storageConfig:
    metricObjectStorage:
      name: thanos-object-storage
      key: thanos.yaml
  retentionResolution1h: 24h
  retentionResolution5m: 24h
  retentionResolutionRaw: 24h

```

1.4 Deploying a VM from the ACM Hub Cluster to ocp-v Using GitOps

With the second cluster imported and virtualization installed, ACM's GitOps integration can now push a VM definition to it automatically. This requires linking ArgoCD to the target cluster through a Placement, then pointing it at the Git repo containing the VM manifest.

- First, define a Placement so ArgoCD knows which managed cluster to target

```

apiVersion: cluster.open-cluster-management.io/v1beta1

```

```

kind: Placement
metadata:
  name: vm-cluster-placement
  namespace: openshift-gitops
spec:
  predicates:
    - requiredClusterSelector:
        labelSelector:
          matchLabels:
            name: ocp-v

```

- Next, bind the default cluster set to the openshift-gitops namespace so ArgoCD is permitted to use it

```

apiVersion: cluster.open-cluster-management.io/v1beta2
kind: ManagedClusterSetBinding
metadata:
  name: default
  namespace: openshift-gitops
spec:
  clusterSet: default

```

- Then create the GitOpsCluster resource, which registers the placement output with ArgoCD as a usable destination

```

apiVersion: apps.open-cluster-management.io/v1beta1
kind: GitOpsCluster
metadata:
  name: vm-gitops-cluster
  namespace: openshift-gitops
spec:
  argoServer:
    cluster: in-cluster
    argoNamespace: openshift-gitops
  placementRef:
    kind: Placement
    apiVersion: cluster.open-cluster-management.io/v1beta1
    name: vm-cluster-placement

```

- Finally, apply the ArgoCD Application below; it syncs the VM manifest from the GitHub repo and deploys it to ocp-v

```

apiVersion: argoproj.io/v1alpha1
kind: Application

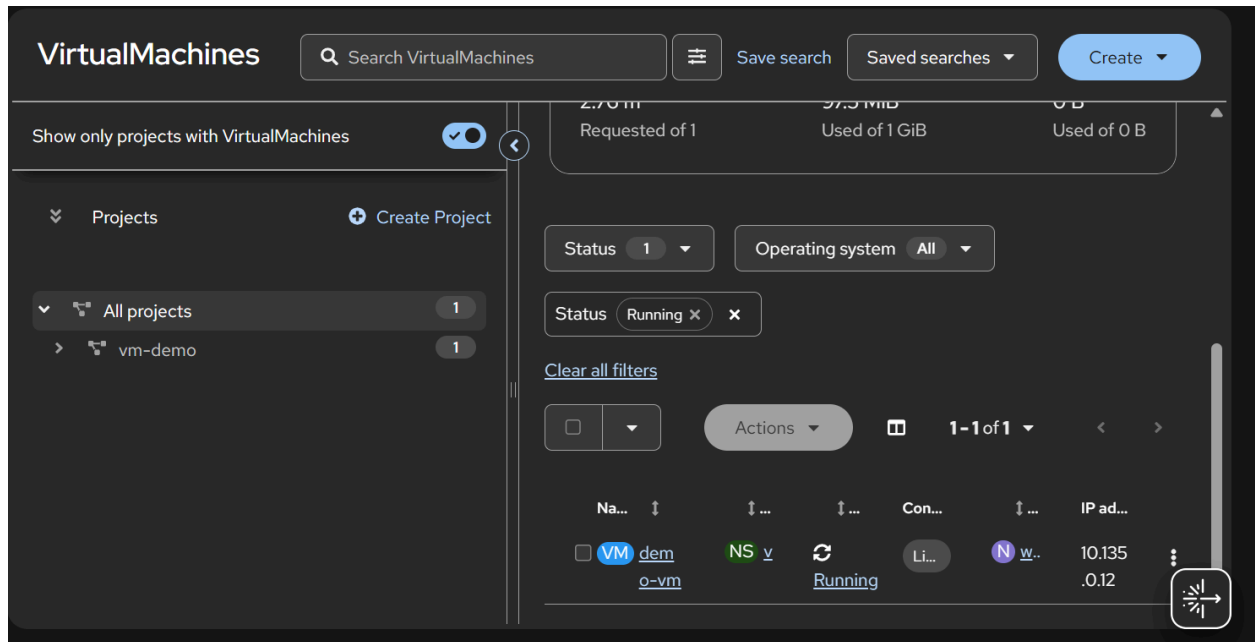
```

```

metadata:
  name: vm-gitops
  namespace: openshift-gitops
spec:
  project: default
  source:
    repoURL: https://github.com/JayarajVp/git_vms.git
    targetRevision: HEAD
    path: .
  destination:
    name: ocp-v
    namespace: vm-demo
  syncPolicy:
    automated:
      prune: true
      selfHeal: true
    syncOptions:
      - CreateNamespace=true

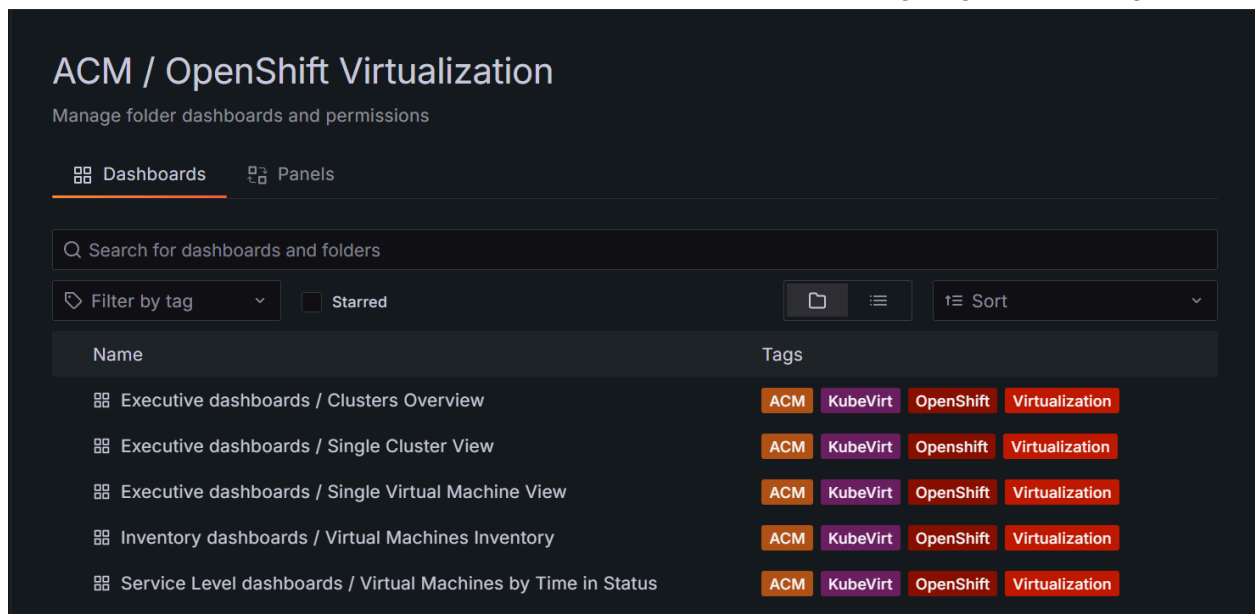
```

The screenshot displays the Argo CD web interface for the 'vm-gitops' application. The left sidebar shows the navigation menu with options like Applications, Settings, User Info, and Documentation. The main content area is titled 'Applications / vm-gitops' and includes a top bar with buttons for DETAILS, DIFF, SYNC, SYNC STATUS, HISTORY AND ROLLBACK, DELETE, and REFRESH. Below this, the 'APP HEALTH' section shows a 'Healthy' status with a green heart icon. The 'SYNC STATUS' section indicates the application is 'Synced to HEAD (dd7193b)' with a green checkmark. The 'LAST SYNC' section shows a successful sync 'Sync OK to dd7193b' that occurred 8 hours ago. The bottom section features a dependency graph showing the application's components: 'vm-gitops' (source) points to 'vm-demo' (namespace), which points to 'demo-vm' (virtualmachine), which in turn points to 'revision-start-vm-84a5832b-c...' (controllerrevision), and finally to 'demo-vm' (virtualmachineinstance). Each component has a status icon and a timestamp indicating when it was last updated.

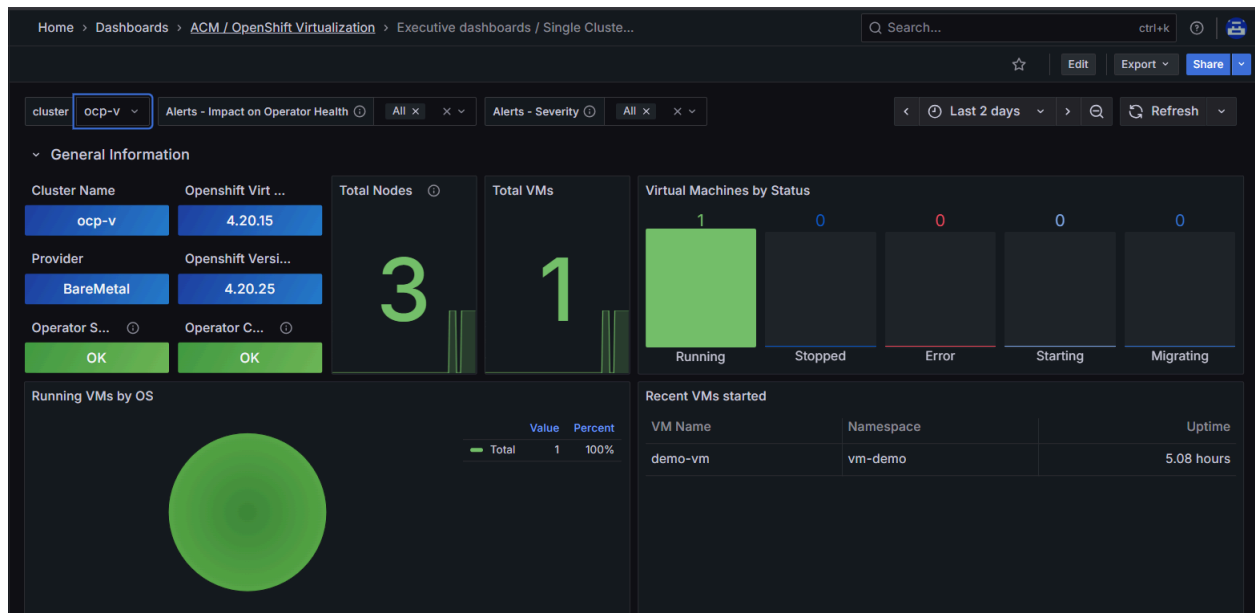
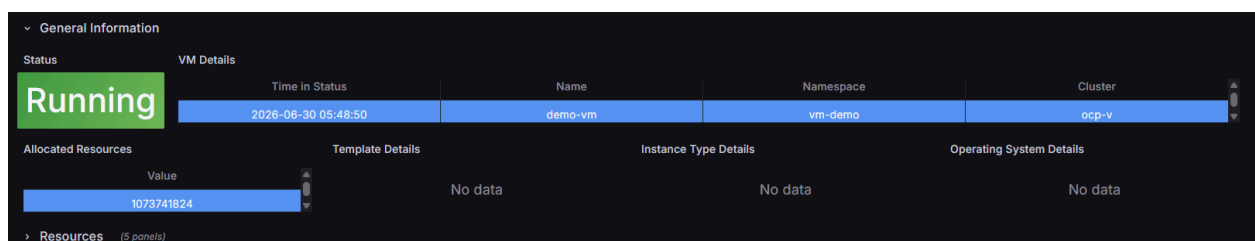
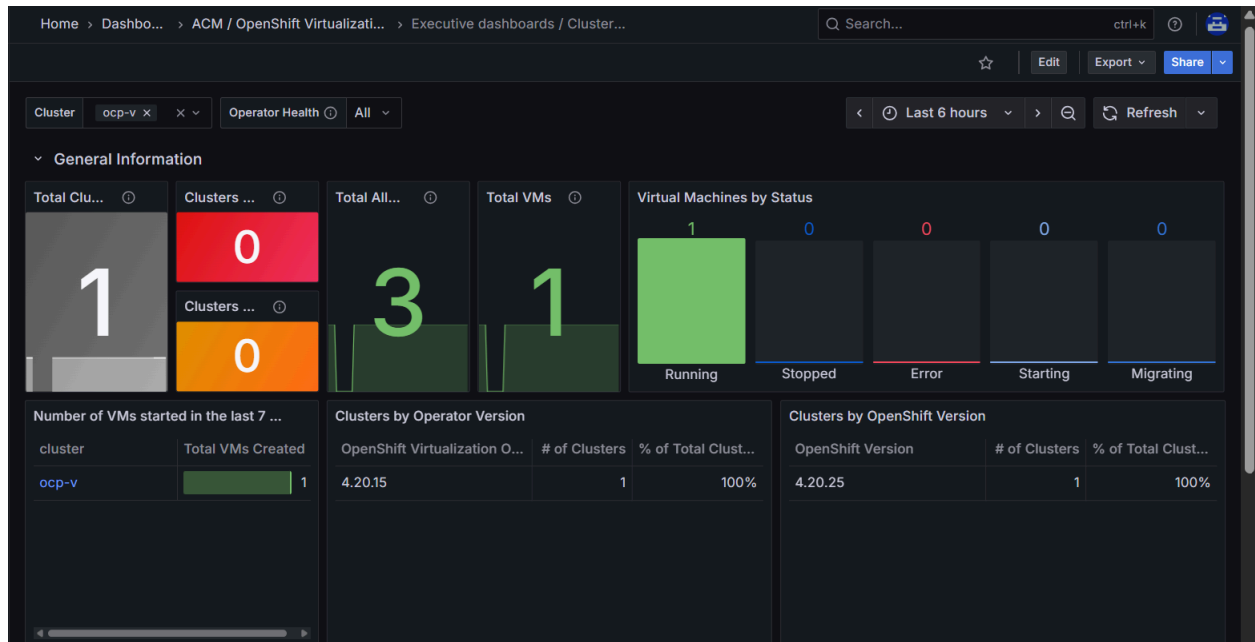


1.5 ACM Grafana Dashboard

ACM ships a built-in Grafana dashboard for OpenShift Virtualization, giving the following views:



General information



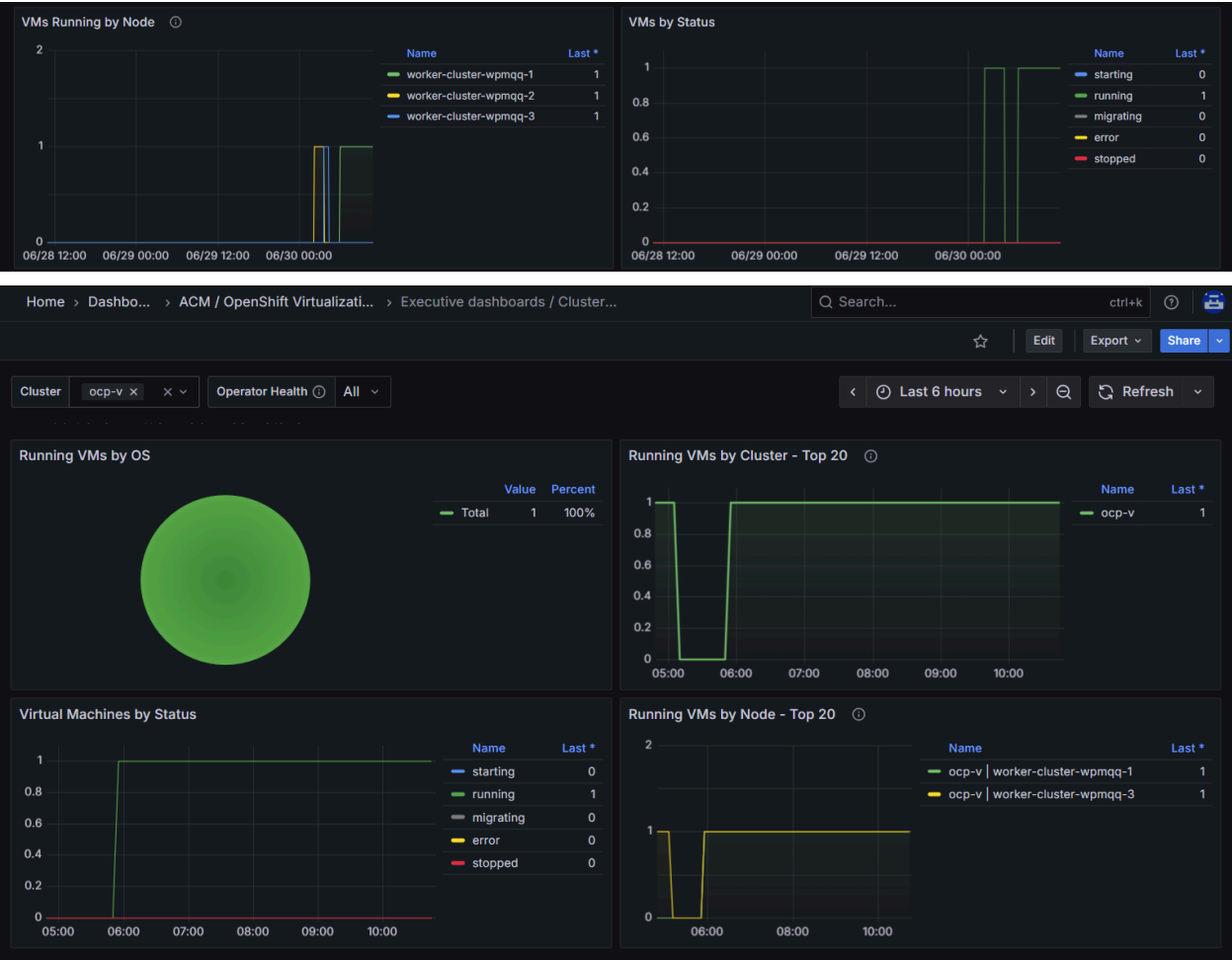
Operator health

Operator Health

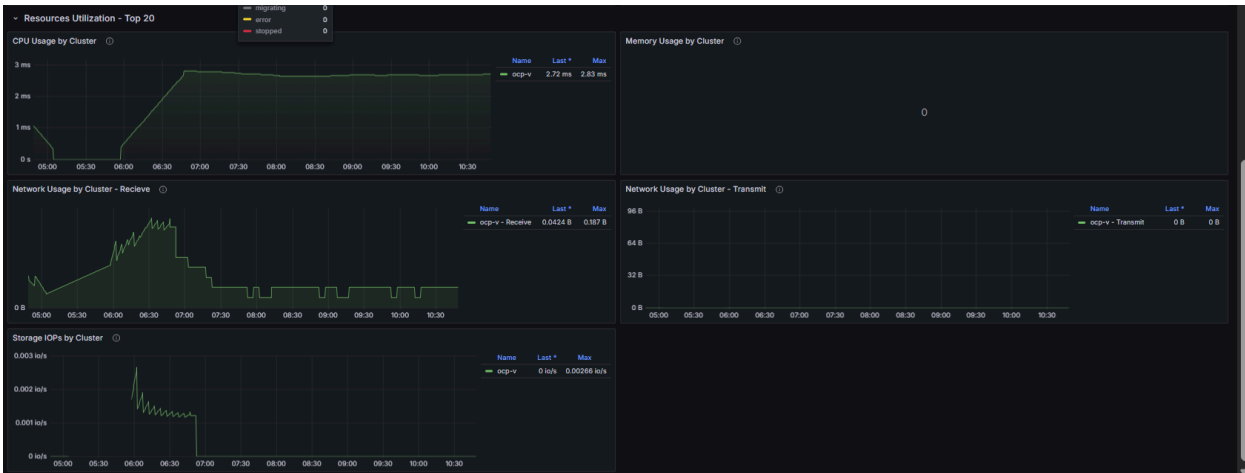
Operator Health by Cluster

Operator Health	Cluster	Operator Conditions Health	Alerts with Critical Impact	Alerts with Warning Impact	Number of Running VMs
Healthy	ocp-v	Healthy	0	0	1

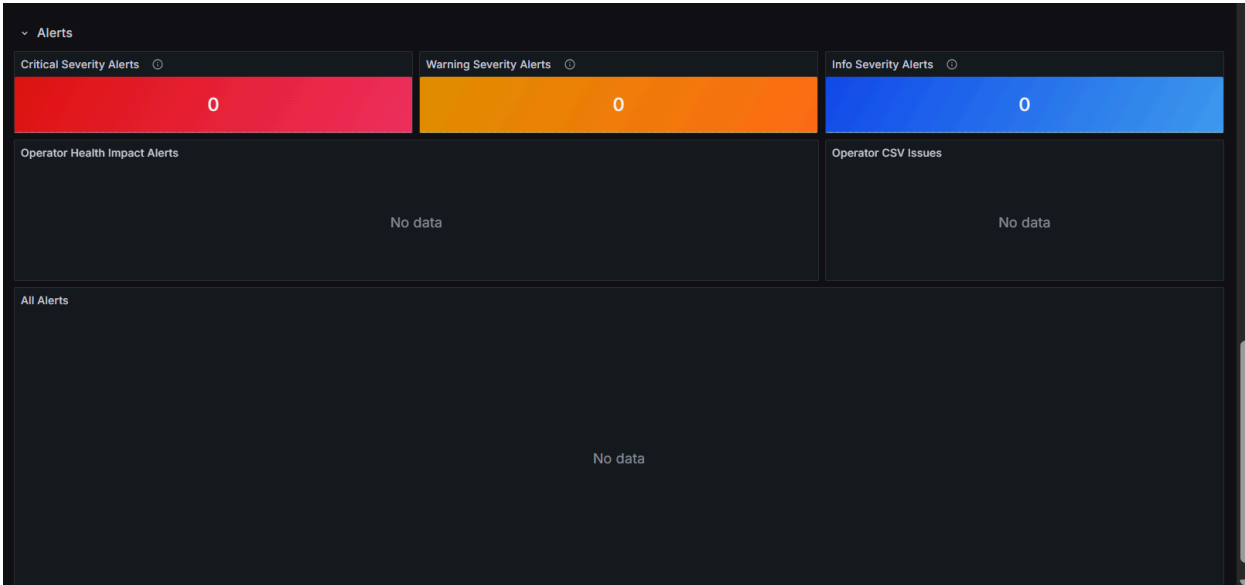
VM status



Resource utilization



VM alerts



VM Inventory

☆EditExportShare

Cluster ocp-vNamespaceAll xVM NameAll x

<Last 6 hours>Refresh

Virtual Machines Inventory

Cluster	Namespace	VM Name	Status	Node	Allocated m... +	Guest Agent	evictable	outdated	vmi_pod
ocp-v	vm-demo	demo-vm	running	worker-cluster-wpmq	1.07 GB	Reporting	true	false	virt-launcher-demo-vi